

Pengping Tan

School of Computer Science and Engineering, Sun Yat-sen University, Guangzhou, China

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EDUCATION

Sun Yat-sen University

M.Eng. in Computer Technology
School of Computer Science and Engineering

Guangzhou, China
Sept. 2024 – Jun. 2027 (expected)

Sun Yat-sen University

B.S. in Information and Computing Science
GPA: 3.8/4.0 | Minor in Psychology

Zhuhai, China
Sept. 2020 – Jun. 2024

RESEARCH INTERESTS

Human-Computer Interaction, Human-Centered AI, Natural Language Processing

RESEARCH PROJECTS

JobMate: AI-Assisted Career Exploration with Real Stories from Social Media

Project Lead | Advisor: Prof. Zhenhui Peng

Guangzhou, China
Oct. 2025 – Apr. 2026

- Initiated the project after identifying limitations in existing approaches to career exploration; worked under Prof. Peng's guidance to focus the research on career stories shared on social media and develop the study design.
- Led the project end to end: conducted a formative study comprising a survey of 64 participants and eight semi-structured interviews, developed the data pipeline and interaction design, and independently built the full-stack JobMate system.
- Designed and ran a controlled mixed-methods user study with 24 participants; conducted the statistical and thematic analyses and led manuscript preparation. Results showed that JobMate helped participants move beyond social comparison to reframe and make sense of their own situations.

Investor Sentiment Analysis and Stock Price Prediction with Integrated Emojis

Team Lead | Faculty Mentor: Prof. Patrick Houlihan, Columbia University

Remote
Jul. 2025 – Sept. 2025

- Led a four-person team investigating whether emoji signals could improve sentiment-aware stock-price prediction; developed the research plan, assigned responsibilities, and delivered biweekly progress presentations.
- Co-designed the LSTM and emoji-fusion components of a model combining FinBERT text embeddings, Emoji2Vec representations, and historical market indicators to predict next-day closing prices for three stocks using StockEmotions, a dataset of 10,000 StockTwits posts.
- Led the literature review and manuscript development; emoji features reduced MSE for all three stocks, most notably Boeing (42.3%), resulting in a co-first-authored ICFTBA 2025 paper.

Bai Juyi's Poetry and His Literary Encyclopedia: A Computational Study

Undergraduate Thesis | Advisors: Prof. Baoquan Zhao & Prof. Zhitian Wang

Zhuhai, China
Nov. 2023 – May 2024

- Extended Prof. Wang's manual study by proposing a computational approach; worked with him to understand the original coding framework and interpretation, and refined the technical plan with guidance from Prof. Zhao.
- Independently built a Python pipeline to parse and organize 40 volumes of Bai Juyi's poetry and extract 20,667 terms and allusions from 30 volumes of *Baishi Liutie*, using rule-based text processing, exact string matching, statistical analysis, and visualization.
- Identified 1,246 exact textual matches and broadly reproduced the distributional trends found in the manual analysis; found matches in 49% of leisure poems and 44% of regulated verse, highlighting the coexistence of plain language and extensive classical references in Bai Juyi's poetry.

PUBLICATIONS

- Tan, P.**, Zhao, B., & Peng, Z. (2026). Same Stories, Different Journeys: From Social Comparison to Sensemaking in AI-Mediated Peer Career Exploration. arXiv preprint; manuscript under revision. <https://arxiv.org/abs/2607.11039>
- Tan, P.**^{*}, Chen, J.^{*}, Wu, R.^{*}, & Liu, H. (2025). Investor Sentiment Analysis and Stock Price Prediction with Integrated Emojis. ICFTBA 2025. ^{*}Equal contribution. <https://doi.org/10.54254/2754-1169/2025.GL28959>

TEACHING EXPERIENCE

Private Mathematics Tutor

High School Mathematics | One-on-One Instruction

China
Jan. 2024 – Jul. 2024

- Tutored nine high school students in weekly two-hour mathematics sessions, preparing individualized problem-solving lessons and adapting explanations to each student's needs and progress in both in-person and online settings.

AWARDS & SCHOLARSHIPS

- Outstanding Student Scholarship**, Sun Yat-sen University: Second Prize (2023 – 2024); Third Prize (2021 – 2023)

SKILLS

- Research Methods:** Surveys, semi-structured interviews, mixed-methods user studies, statistical and thematic analysis
- Technical Skills:** Python, PyTorch, JavaScript, LLM APIs, Vue.js, Node.js, MongoDB, Neo4j
- Languages:** English (IELTS Academic: 7.0), Mandarin Chinese (native)